



Introduction to Cosmology and Data Driven Analysis of Λ CDM Model

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SXPC-NEPAL

Subhash Bhandari¹

Department of Physics

St. Xavier's College, Maitighar, Kathmandu, Nepal

¹cosubhasmos@gmail.com

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1 Introduction

This is an introductory lecture notes on cosmology aimed at participants of the SXPC Workshop on Astrophysics and Cosmology, 2026. Throughout this note, I will introduce you to the basics of cosmology, especially focusing on the standard Λ CDM model, followed by a hands-on analysis of its dynamical stability by constructing a phase space diagram. We will wrap up the workshop with a simple MCMC analysis and fitting with the Hubble dataset. You will need to be comfortable with the basics of Special Relativity, but no prior knowledge of either General Relativity or Statistical Mechanics is assumed. The diagrams and plots in these notes are generated using the code provided in the [Github](#).

1.1 Overview of Cosmology

If you have read basic physics in your high school, you might have probably heard about the Big Bang Theory. So what actually is the Big Bang theory? You might answer: simple, it's a theory that explains the origin of the universe. Wrong! The Big Bang theory is not about the origin of the universe. Then you might ask, then how did the universe begin if it's not the Big Bang? We don't know. The Big Bang Theory has a slight misinterpretation when it is read. People think it is a theory that says like it went bang² and then the universe began. But the Big Bang theory is all about the evolution of the universe, and was never about the birth of the universe.

According to the Big Bang theory, there was a time at the very beginning, when the Universe was so hot that matter, atoms, and even nuclei melted and all of space was filled with a fireball³. The Big Bang theory is a collection of ideas, calculations, and predictions that explain what happened in this fireball, and how it subsequently evolved into the universe we see around us today. We actually have proof of this fireball thing, and in a fancy name, we call it Cosmic Microwave Background Radiation, as shown in the Figure 1. Actually, David Tong⁴ has his own

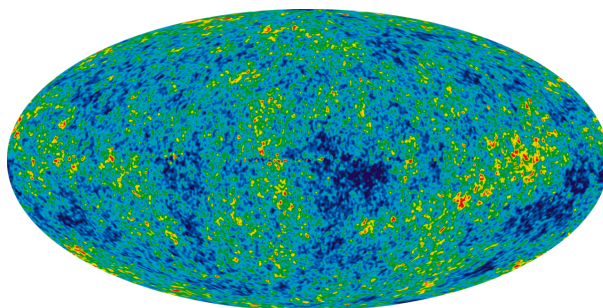


Figure 1: Cosmic microwave radiation released approximately 375,000 years after the birth of the universe. Image: NASA / WMAP Science Team

perspective about this fancy name[18].

Of course, this being science, we don't like to brag about these things, so rather than jumping up and down and shouting "we've taken a fucking photograph of the fucking Big Bang", we instead wrap it up in dull technical words. We call it the cosmic microwave background radiation.

Now, the goal of the physicists is to go as far as possible into the past of the universe and understand what actually happened. We call it cosmology!

²biiiiig one

³a ball of fire

⁴one of my favorite physicists

1.2 Cosmological Principle

Now, let's start with some basics of cosmology. The cosmological principle is a foundational assumption in modern cosmology, stating that the universe is homogenous and isotropic on large scales⁵. By homogenous, we mean that the distribution of matter and energy is uniform throughout the cosmos when averaged over very large scales, and by isotropic, it means that the universe looks the same no matter which direction one looks at, or simply we are not in a special place of the universe. The cosmological principle is a hypothesis that actually makes the very large portion of cosmological theory. Again, Cosmic Microwave Background Radiation is good evidence of this principle. In more mathematical language, the Cosmological Principle can be formulated as [20],

- The hypersurfaces with constant cosmic standard time are maximally symmetric subspaces of the whole of space-time.
- Not only the metric $g_{\mu\nu}$, but all cosmic tensors such as $T_{\mu\nu}$ are form-invariant with respect to the isometries of these subspaces.

1.3 The Robertson-Walker Metric

We apply the formulation from the Cosmological Principle to spaces with maximally symmetric subspaces. From which it must be possible to choose coordinates r, θ, ϕ, t , then A 4-D manifold with this symmetry must possess a maximally spatially symmetric FRW metric,

$$ds^2 = -dt^2 + a(t)^2 \left(\frac{dr^2}{1 - \kappa r^2} + r^2 d\theta^2 + r^2 \sin \theta d\phi^2 \right) \quad (1)$$

where κ is a spatial curvature term with a value of $(-1, 0, +1)$ for an open, flat, and closed universe, respectively. $a(t)$ is a scale factor that is a function of the time coordinate. The scale factor $a(t)$ is linearly related to spatial lengths in the universe.

The scale factor $a(t)$ is linearly related to the spatial length of the universe. Let us take a simple example. The radial distance between two points A and B in the universe is

$$L_{AB} = \int_A^B ds = a(t) \int_A^B \frac{dr}{1 - \kappa r^2} \quad (2)$$

This equation shows how the separation between two points in the universe evolves in time due to the presence of the scale factor. In other words, the larger the scale factor, the farther away there will be points in the universe. For a flat universe, $\kappa = 0$ and hence $L_{AB} = a(t)(r_B - r_A)$.

2 Dark Energy

The nature of dark energy is a pivotal question in cosmology, with implications for fundamental physics and the universe's fate. The discovery of the universe's accelerated expansion in the late 1990s, through observations of Type Ia supernovae [15, 11], introduced dark energy as a dominant component driving this phenomenon. Dark energy constitutes approximately 68% of the universe's energy in the Λ CDM model, and exerts negative pressure, counteracting gravitational collapse and accelerating cosmic expansion [12]. Despite its critical role, the physical nature of dark energy remains one of cosmology's greatest mysteries, with candidates ranging from a cosmological constant to dynamical scalar fields [5].

The cosmological constant, Λ , characterized by an equation of state (EOS) $w = -1$, is the simplest and so-called standard model of cosmology to date, but faces some theoretical

⁵by large scale, we mean the cells of diameter 10^8 to 10^9 light years

challenges, such as the fine-tuning problem [21]. Alternative models propose a time-varying EOS, $w(z)$, where z is the redshift, allowing dark energy to evolve dynamically. These include quintessence [13, 4], phantom fields [3], and quintom models [7], which permit $w(z)$ to cross the $w = -1$ boundary. Observational constraints on $w(z)$ are crucial to distinguish between these models and test deviations from Λ CDM.

Recent studies have leveraged diverse cosmological probes to constrain dark energy. The Pantheon sample of Type Ia supernovae [16] provides precise distance measurements, while baryon acoustic oscillations (BAO) from the Sloan Digital Sky Survey (SDSS) and DESI [2] map the universe's expansion history. Hubble parameter measurements from cosmic chronometers [10] offer model-independent constraints. Notably, the DESI 2024 data suggest a preference for dynamical dark energy, with evidence of a quintom-like EOS crossing from phantom ($w < -1$) to quintessence ($w > -1$) regimes [22, 19]. These findings motivate further exploration of dynamical models to address cosmological tensions, such as the Hubble tension [14].

3 The Λ Driven Dark Energy

The Λ CDM model is also called the standard model of cosmology for its simplicity and validity. The Λ represents the cosmological constant (dark energy) causing accelerated expansion, while CDM stands for Cold Dark Matter, which provides the gravitational scaffolding for structure formation. This universe follows the Big Bang hypotheses followed by a rapid expansion called inflation that created large-scale uniformity. The model successfully predicts the Cosmic Microwave Background (CMB), Big Bang nucleosynthesis (light element abundance), and Large-Scale Structure (LSS). While being heavily supported by data from, for example, the Planck satellite, which mapped the CMB, as well as Baryon Acoustic Oscillations (BAO) and supernova surveys, this model faces some challenges, including Hubble Tension, and recent signatures of dynamical dark energy [1]. Despite the discrepancies, this model is older than many modern astrophysicists realize, and it has not fundamentally changed for over twenty-five years[17].

The Λ CDM model, while successful, assumes a constant EOS, which may not capture potential dynamical behavior suggested by recent data. Parameterizing $w(z)$ allows us to probe deviations from $w = -1$, offering insights into dark energy's evolution. The Chevallier-Polarski-Linder (CPL) parameterization, $w(a) = w_0 + w_a(1 - a)$, where $a = 1/(1 + z)$ is the scale factor, is widely used but limited in capturing complex dynamics [6, 9]. Models like the quintom, which allow $w(z)$ to cross the cosmological constant boundary, provide a more flexible framework [7]. This paper focuses on constraining dark energy parameters and $w(z)$ using a quintom model, combining Hubble, Pantheon+, and DESI BAO data. We aim to test the model's consistency with observations, assess its dynamical stability, and explore its implications for resolving cosmological tensions.

3.1 Einstein's Equation

The dynamics of metric tensor $g_{\mu\nu}$ is given by Einstein Field equation,

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} = \kappa^2 T_{\mu\nu} \quad (3)$$

where $R = g^{\mu\nu}$ is Ricci or curvature scalar. For a perfect fluid we can describe the matter inside the homogeneous and isotropic universe at large scales with good precision and its energy momentum tensor can be written as,

$$T_{\mu\nu} = pg_{\mu\nu} + (\rho + p)u_\mu u_\nu \quad (4)$$

where, u^μ is a 4-velocity vector of an observer co-moving with fluid, $u^\mu = (-1, 0, 0, 0)$.

Equation of state relates the energy density and pressure and is written as,

$$p = \omega \rho \quad (5)$$

For perfect fluid, $\omega = 0$ and for relativistic fluid $\omega = 1/3$.

3.2 Cosmological Equations

After solving Einstein Field Equations 3 by the use of FRW metric 1, the equation arises in two coupled differential forms for $a(t)$ and matter variables $\rho(t)$ and $p(t)$.

$$\frac{\kappa}{a^2} + H^2 = \frac{\kappa^2}{3} \rho \quad (6)$$

$$\frac{\kappa}{a^2} + 2\dot{H} + 3H^2 = -\kappa^2 p \quad (7)$$

Using Friedmann Equation 6 and acceleration Equation 7,

$$\frac{\ddot{a}}{a} = -\frac{\kappa^2}{6}(\rho + 3p) \quad (8)$$

If $\rho + 3p > 0$ the universe is decelerating and if $\rho + 3p < 0$ then the universe is said to be accelerating.

3.3 Energy Conservation Equations

From the conservation of electromagnetic tensor $\nabla_\mu T^{\mu\nu}$ or equivalently from the Friedmann and acceleration equation, gives the continuity equation,

$$\dot{\rho} + 3H(\rho + p) = 0 \quad (9)$$

From linear equation of state 5, the above equation gives the solution,

$$\rho \propto a^{-3(\omega+1)} \quad (10)$$

For dust like fluid, $p_m = 0$, and for radiation $p_r = \rho_r/3$, then solving for Equation 10 gives,

$$\rho_m \propto a^{-3} \quad \rho_r \propto a^{-4} \quad (11)$$

If we consider multiple fluid sources in the universe, then the conservation equation of the electromagnetic tensor becomes,

$$\nabla^\mu T_{\mu\nu} = \nabla^\mu (T_{\mu\nu}^{(1)} + T_{\mu\nu}^{(2)}) = 0 \quad (12)$$

Since the fluid sources are interacting with each other, the individual term for $\nabla_\mu T^{\mu\nu}$ cannot be zero. Instead they will take the form $\nabla_\mu T_{\mu\nu}^{(1)} = Q_\nu$ and $\nabla_\mu T_{\mu\nu}^{(2)} = -Q_\nu$ ending up zero as a whole.

3.4 Cosmological Constant

The cosmological constant was first introduced by Einstein in his field equations 3, and can be rewritten as,

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa^2 T_{\mu\nu} \quad (13)$$

where Λ is a cosmological constant. Again considering FRW line element 1 and new field equations 13 we get our new cosmological equations,

$$3H^2 = \kappa^2 \rho + \Lambda \quad (14)$$

$$2\dot{H} + 3H^2 = -\kappa^2 p + \Lambda \quad (15)$$

The EoS parameter of the cosmological constant has the value of $\omega_\Lambda = -1$ since $p_\Lambda = -\rho_\Lambda$. Consider the cosmological constants with both matter and radiation for which the values, ($p_m = 0$) and ($p_r = \rho_r/3$), equation 14 takes the form,

$$3H^2 = \kappa^2 \rho_m + \kappa^2 \rho_r + \Lambda \quad (16)$$

$$2\dot{H} + 3H^2 = -\frac{\kappa^2}{3} \rho_r + \Lambda \quad (17)$$

Let us now define the dimensionless variables from the Friedmann equations 6,

$$x = \Omega_m = \frac{\kappa^2 \rho_m}{3H^2} \quad y = \Omega_r = \frac{\kappa^2 \rho_r}{3H^2} \quad (18)$$

Using these variables in cosmological equations, we can obtain the Friedmann equation in the form,

$$1 = x + y + \Omega_\Lambda \quad (19)$$

where $\Omega_\Lambda = \Lambda/3H^2$ is the relative energy density of the cosmological constant, or simply dark energy in our case.

3.5 Dynamical System and Physical Phase Space

At some point in the universe, during the transition from radiation to matter, the energy of matter will be higher and will dominate the universe afterwards. In this model, we assume the interaction between matter and radiation to be zero, but the scale factor is said to be evolving with time.

To obtain a dynamical system, we differentiate variables x and y from equation 18 with respect to $\log a$. We have from equation 18,

$$x = \frac{\kappa^2 \rho_m}{3H^2}$$

Now,

$$\begin{aligned} \frac{dx}{dt} &= \frac{d}{dt} \left(\frac{\kappa^2 \rho_m}{3H^2} \right) \\ &= \frac{\kappa^2}{3} \left[\frac{\dot{\rho}_m}{H^2} + \frac{2\rho_m}{H^2} \frac{\dot{H}}{H} \right] \\ \frac{1}{H} \frac{dx}{dt} &= \frac{\kappa^2}{3} \left[\frac{\dot{\rho}_m}{H^3} + \frac{2\rho_m}{H^2} \frac{\dot{H}}{H} \right] \end{aligned}$$

Thus,

$$x' = \frac{1}{H} \frac{dx}{dt} = \frac{\kappa^2 \dot{\rho}_m}{3H^3} - \frac{2\kappa^2 \rho_m}{3H^2} \frac{\dot{H}}{H^2} \quad (20)$$

Since we have already assumed no interaction between matter and radiation, from conservation equation for two fluids 9 we get

$$\dot{\rho}_m = -3H\rho_m, \quad \dot{\rho}_r = -4H\rho_r \quad (21)$$

where, $p_m = 0$ and $p_r = 1/3$ are assumed. The acceleration equation 16 gives,

$$\frac{\dot{H}}{H^2} = -\frac{3}{2} - \frac{\kappa^2 \rho_r}{6H^2} + \frac{\Lambda}{2H^2} = -\frac{1}{2}(3 + y - 3\Omega_\Lambda) \quad (22)$$

Also, from Friedmann equation 19, we have

$$\Omega_\Lambda = 1 - x - y \quad (23)$$

Thus, the equation 22 becomes,

$$\begin{aligned} \frac{\dot{H}}{H^2} &= -\frac{1}{2}(3 + y - 3(1 - x - y)) \\ &= -\frac{1}{2}(3 + y - 3 + 3x + 3y) \\ &= -\frac{1}{2}(4y + 3x) \end{aligned}$$

Thus,

$$\boxed{\frac{\dot{H}}{H^2} = -\frac{1}{2}(4y + 3x)} \quad (24)$$

Substituting these results in x' and y' we obtain,

$$\begin{aligned} x' &= \frac{\kappa^2 \dot{\rho}_m}{3H^3} - \frac{2\kappa^2 \rho_m}{3H^2} \frac{\dot{H}}{H} \\ &= \frac{\kappa^2}{3H^3}(-3H\rho_m) - \frac{2\kappa^2 \rho_m}{3H^2} \left[-\frac{1}{2}(4y + 3x) \right] \\ &= -3\frac{\kappa^2 \rho_m}{3H^2} - 2\frac{\kappa^2 \rho_m}{3H^2} \left[-\frac{1}{2}(4y + 3x) \right] \\ &= -3x - 2x(-2y - \frac{3}{2}x) \\ &= -3x + 4xy + 3x^2 \\ &= x(3x + 4y - 3) \end{aligned}$$

Similarly, we have from equation, 18

$$y = \frac{\kappa^2 \rho_r}{3H^2}$$

Now,

$$\begin{aligned} \frac{dy}{dt} &= \frac{d}{dt} \left(\frac{\kappa^2 \rho_r}{3H^2} \right) \\ &= \frac{\kappa^2}{3} \left[\frac{\dot{\rho}_r}{H^2} + \frac{2\rho_r}{H^2} \frac{\dot{H}}{H} \right] \\ \frac{1}{H} \frac{dy}{dt} &= \frac{\kappa^2}{3} \left[\frac{\dot{\rho}_r}{H^3} + \frac{2\rho_r}{H^2} \frac{\dot{H}}{H} \right] \end{aligned}$$

Thus,

$$\boxed{y' = \frac{1}{H} \frac{dy}{dt} = \frac{\kappa^2 \dot{\rho}_r}{3H^3} - \frac{2\kappa^2 \rho_r}{3H^2} \frac{\dot{H}}{H^2}} \quad (25)$$

Again, from conservation equation 9 and result from Friedmann equation 24, we have

$$\dot{\rho}_r = -4H\rho_r \quad \frac{\dot{H}}{H^2} = -\frac{1}{2}(4y + 3x)$$

Substituting these values in equation 25,

$$\begin{aligned}
y' &= \frac{\kappa^2 \dot{\rho}_r}{3H^3} - \frac{2\kappa^2 \rho_r}{3H^2} \frac{\dot{H}}{H^2} \\
&= \frac{\kappa^2}{3H^3} (-4H\rho_r) - \frac{2\kappa^2 \rho_r}{3H^2} \left[-\frac{1}{2}(4y + 3x) \right] \\
&= -4 \frac{\kappa^2 \rho_r}{3H^2} - 2 \frac{\kappa^2 \rho_r}{3H^2} \left[-\frac{1}{2}(4y + 3x) \right] \\
&= -4y - 2y(-2y - \frac{3}{2}x) \\
&= -4y + 4y^2 + 3xy \\
&= y(3x + 4y - 4)
\end{aligned}$$

Hence, we obtain our dynamical system as,

$$x' = x(3x + 4y - 3) \quad y' = y(3x + 4y - 4) \quad (26)$$

Now, we can obtain critical points of the system by equating equations 26 to zero, $x' = 0$ and $y' = 0$, we get two points $x = 0$ and $y = 0$. Now

$$\text{For, } y = 0 \quad x' = 0 \quad \implies 3x + 4 \cdot 0 - 3 = 0 \quad \implies x = 1 \quad (27)$$

$$\text{For, } x = 0 \quad y' = 0 \quad \implies 3 \cdot 0 + 4y - 4 = 0 \quad \implies y = 1 \quad (28)$$

This will lead to critical points $O(0,0)$, $R(0,1)$, and $M(1,0)$, and after performing stability analysis, we obtain the eigenvalues as shown in Table 1. The effective EoS parameter w_{eff} is given by

$$\begin{aligned}
\omega_{eff} &= \omega_m \Omega_m + \omega_R \Omega_R + \omega_\Lambda \Omega_\Lambda \\
&= 0 \cdot \Omega_m + \frac{1}{3} \cdot \Omega_R + (-1) \cdot \Omega_\Lambda
\end{aligned}$$

From Friedmann equation 19 we have $\Omega_\Lambda = 1 - x - y$, and from our assumption $\Omega_R = y$, thus

$$\begin{aligned}
\omega_{eff} &= \frac{1}{3} \cdot y + (-1) \cdot (1 - x - y) \\
&= \frac{1}{3}y - 1 + x + y \\
&= -1 + x + \frac{4}{3}y \\
\boxed{\omega_{eff} = -1 + x + 4y/3} & \quad (29)
\end{aligned}$$

, and takes the values for different critical points x and y as shown in the same table. From

Point	x	y	ω_{eff}	Eigenvalues	Stability
O	0	0	-1	(-4,-3)	Stable
R	0	1	1/3	(1,4)	Unstable
M	1	0	0	(-1,3)	Saddle

Table 1: Critical points and eigenvalues of the system for Λ CDM Model

Table 1, we can clearly see that the value of w_{eff} for a cosmological constant-dominated universe is -1 , for a radiation-dominated universe (point R) is $1/3$, and for matter-dominated universe (point M) is 0 .

3.6 Phase Space Diagram

The physical phase space of the system has been plotted in Figure 2. The point O corresponds to a cosmologically constant-dominated universe or a dark energy-dominated universe, R corresponds to a radiation-dominated universe, and M corresponds to a matter-dominated universe. In the figure, point R is clearly the past attractor while origin O represents the future attractor. Hence, every solution makes a trajectory starting from R and ending in O except the one on the x-axis and the line $y = 1 - x$, which corresponds to either a vanishing cosmological constant or to a universe without a radiation contribution. The light yellow/shaded region in the figure denotes the area of the phase space where $\omega_{eff} < -1/3$, i.e., where the universe undergoes an accelerated expansion. All the trajectories will eventually enter this region (shaded region) so that the radiation (or matter) to dark energy transition always happens at some moment in the history of the universe.

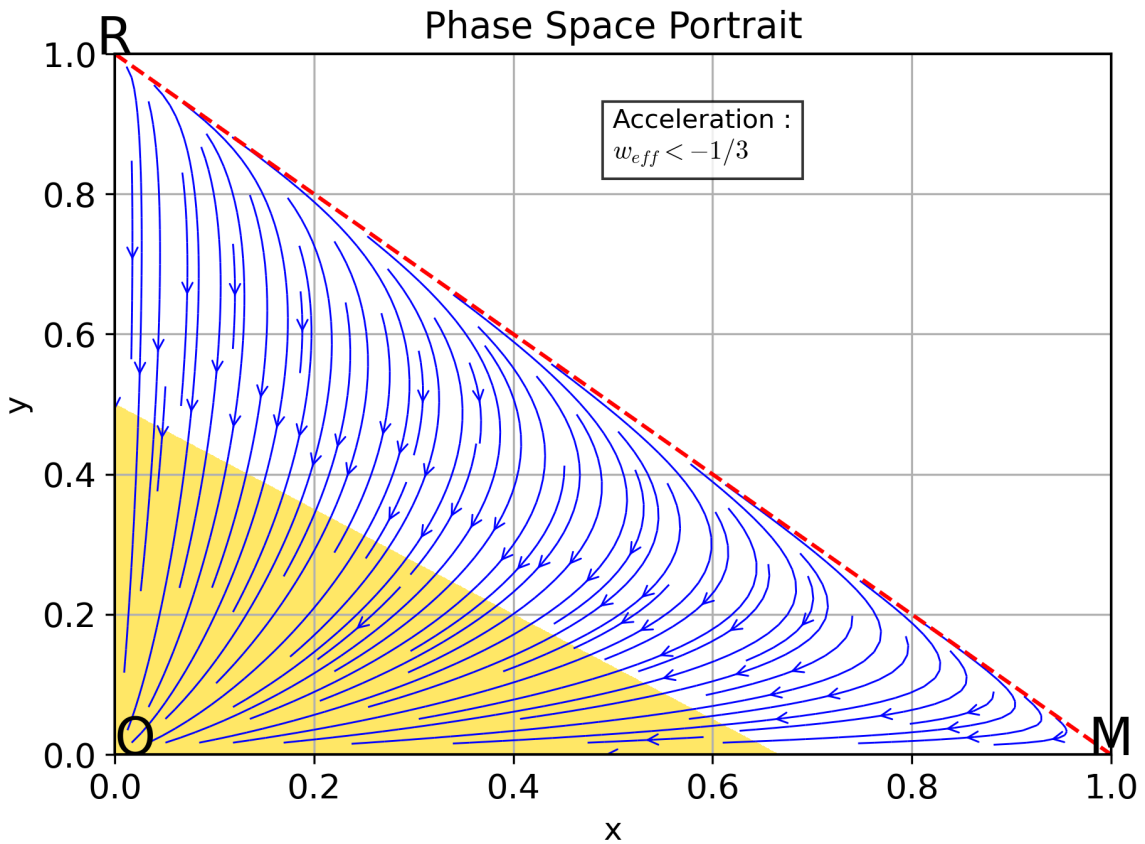


Figure 2: Phase space portrait of Λ CDM Model

4 Constraining Λ CDM Parameters

4.1 Model Fitting: Bayesian Inference with MCMC

There are two different ways of interpreting probability. In the classical notion, the probability is the frequency of recurring events. In a classical or “frequentist” approach, probabilities are objective and tied to the idea of repeatable experiments. We imagine that the true parameters of the system are fixed but unknown, and our data are random outcomes of repeated experiments governed by these parameters. On the other hand, the Bayesian world interprets probability as

the degree of belief or subjective certainty about the truth of the proposition. This theory is more general than classical or frequentist theory.

4.2 Bayesian Framework

In a Bayesian world, uncertainty is about our knowledge of the parameters; hence, the general rule is to update the knowledge of the quantity, in general, from the prior to the posterior. Here, parameters are treated as random variables with prior distributions reflecting our knowledge (or uncertainty) before observing the data. Observing data updates this knowledge via Bayes' theorem. The posterior from a previous cycle of observations becomes the prior for the next. The price we have to pay is that we have to start somewhere by specifying an initial prior, which is not determined by the theory, but it needs to be given by the user. The prior should represent fairly represent the state of knowledge of the user about the quantity of interest. Eventually, the posterior will converge to a unique (objective) result even if different scientists start from different priors (provided their priors are non-zero in regions of parameter space where the likelihood is large).

The joint probability of the events A and B, such that A has already occurred, is given by

$$P(A, B) = P(A)P(B|A) = P(B)P(A|B) \quad (30)$$

from which the statement of Bayes' theorem,

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)} \quad (31)$$

In our scenario, we can infer cosmological parameters by replacing in Bayes theorem, Eq.31, $A \rightarrow \theta$ where $\theta = \{\Omega_M, H_0\}$, the cosmological parameters and $B \rightarrow D$, the observed data, to obtain,

$$P(\theta|D) = \frac{P(D|\theta)P(\theta)}{P(D)} \quad (32)$$

In Equation 32, $P(\theta|D)$ is the posterior probability for θ , and it represents our degree of knowledge about the value of parameters θ after we have seen the data D. Also, $P(D|\theta) = \mathcal{L}(\theta)$ is the likelihood function, $\mathcal{L}(\theta) = P(X = \hat{x}|\theta)$, where the probability of observing the data that have been obtained ($X = \hat{x}$) is considered as a function of the parameters θ . Again, in Equation 32, the quantity $P(\theta)$ is called prior, and represents our degree of belief in the value of parameters before we see the data. Lastly, $P(D)$ is called the normalizing constant, which ensures that the posterior is normalized to unity,

$$P(D) = \sum_{\theta} P(D|\theta)P(\theta) \quad \text{or,} \quad \int P(D|\theta)P(\theta)d\theta \quad (33)$$

Now, if we have the data of Hubble constant H_{obs} and the data returned by the model H_{model} , then the maximized log likelihood is given by

$$\ln \mathcal{L} = -\frac{1}{2} \sum_i \left(\frac{(H_{obs,i} - H_{model,i})^2}{\sigma^2} + \ln(2\pi\sigma^2) \right). \quad (34)$$

In Bayesian inference, we combine this likelihood with the prior to compute the posterior, which tells us how probable each parameter (θ) is, given both the data and prior knowledge.

4.3 MCMC Implementation Using Emcee

Markov Chain Monte Carlo, or shortly MCMC, is a class of algorithms that is constructed to sample from the posterior distribution or a sequence of points in parameter space when the problem cannot be computed analytically. It uses random sampling methods to approximate a quantity by running chains, known as a Markov chain. A Markov chain is a sequence of random variables such that the probability of the next element in the chain only depends on the value of the current element and not the whole history. In the chain at a new step, the proposal distribution suggests a new candidate point of parameter given the current state, and this proposal is either accepted or rejected based on the ratio of posteriors.

There are various algorithms built to solve problems based on their complexity level. Here, we are restricted to the Stretch-Move algorithm because we are using it and nothing else. In this project, we use a popular Python package that uses the stretch-move algorithm, called `emcee`. `emcee` is a Python implementation of the affine-invariant ensemble sampler for Markov chain Monte Carlo (MCMC) proposed by Goodman & Weare [8].

4.4 Posterior Analysis

After running the MCMC sampler, the chains are analysed to extract posterior distributions. We discard the first 10 steps as burn-in to remove initial transients and thin the chains (e.g., every 2nd sample) to mitigate autocorrelation, yielding independent samples from the posterior. Medians and 68% credible intervals are computed as:

$$\theta_{\text{median}}, \quad \sigma^+ = \text{percentile}(84\%) - \theta_{\text{median}}, \quad \sigma^- = \theta_{\text{median}} - \text{percentile}(16\%).$$

4.5 Results

Figure 3 shows the corner plot of the cosmological constraints obtained from fitting the Λ CDM model to the Hubble data. Top-left and bottom-right histograms display the marginalized 1D probability distributions for the matter density parameter Ω_M and the Hubble constant H_0 , respectively. The bottom-left panel is quite an important one. It shows a 2D contour plot showing the joint likelihood of Ω_M and the Hubble constant H_0 . The fit constrains the values, $\Omega_M = 0.26^{+0.02}_{-0.02}$ and $H_0 = 70.60^{+1.42}_{-1.45}$ respectively. The obtained values are consistent with Planck 2018 and SH0ES results, illustrating the Hubble tension when using different datasets. OHD measurements are model-independent probes of expansion history.

Additional Resources

All the resources used in these notes are available on the author's [GitHub Repository](#). For more cosmology codes and dynamical stability analysis of scalar field cosmological models, please visit another repository [here](#). If you have any queries or suggestions, you can contact the author directly via multiple [channels](#).

5 Conclusion and Need for Scalar Field Models

The standard model (Λ CDM), while remarkably successful in explaining a broad array of cosmological observations such as the cosmic microwave background (CMB) anisotropies, large-scale structure formation, and the accelerated expansion of the universe, faces mounting challenges from persistent observational tensions that suggest the need for extensions or alternatives, particularly those involving scalar field dynamics to model dark energy more flexibly. Chief among these is the Hubble tension, a discrepancy exceeding 5σ between local measurements of the Hubble constant H_0 (around 73 km/s/Mpc from Cepheid-calibrated supernovae)

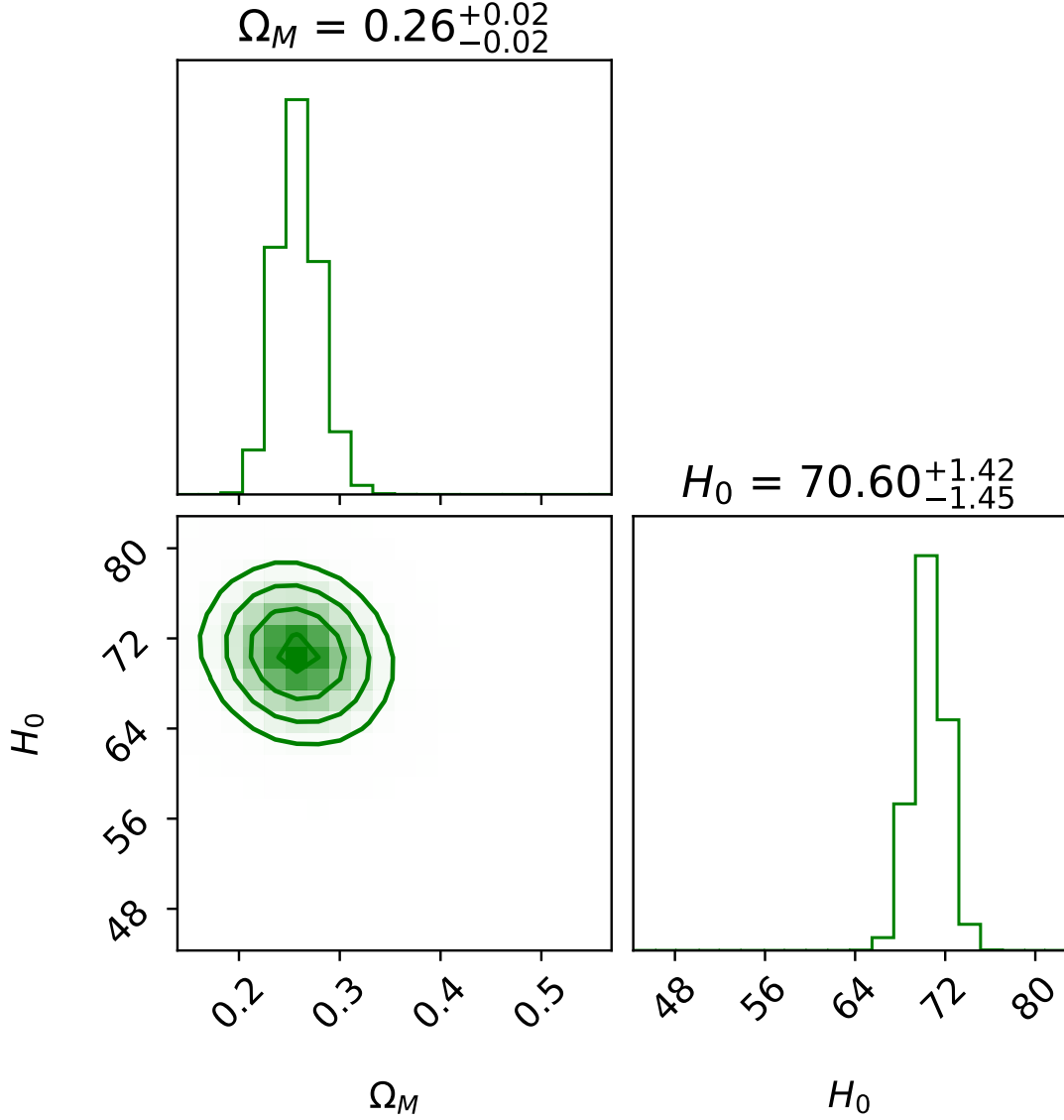


Figure 3: Corner plot showing cosmological parameter constraints from Hubble data

and early-universe inferences from CMB data (around 67 km/s/Mpc under Λ CDM), which implies potential redshift-dependent physics or systematic errors unaccounted for in the standard paradigm. Similarly, the S_8 tension highlights inconsistencies in the amplitude of matter fluctuations, where large-scale structure surveys like KiDS-1000 predict lower values ($S_8 \approx 0.76$) compared to Planck CMB estimates ($S_8 \approx 0.83$), pointing to issues with structure growth, dark matter clustering, or possibly non-gravitational interactions in the dark sector. Recent data from the Dark Energy Spectroscopic Instrument (DESI) further exacerbates these concerns by suggesting a time-varying dark energy equation of state, deviating from the constant $\omega = -1$ of the cosmological constant Λ at up to 3.9σ significance when combined with supernova and CMB datasets, alongside hints of evolving dark energy density that challenge the model's flat geometry and inflation-induced hierarchy problems. Additional anomalies include CMB lensing excesses, curvature constraints, and neutrino mass bounds that collectively indicate "cracks" in Λ CDM, potentially resolvable through new physics like additional relativistic species or modified gravity. These tensions underscore the necessity for scalar field models, such as quintessence or k-essence, where dark energy arises from a dynamical scalar field ϕ with a potential $V(\phi)$, allow-

ing for evolving $\omega(z)$ that can alleviate the Hubble and S_8 discrepancies by adjusting expansion history and perturbation growth without ad hoc parameters.

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